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AN AUTOMATIC BRAIN TUMOR PREDICTION SYSTEM USING TEXTURE PATTERN RECOGNITION APPROACH WITH MATRIX AND CLUSTERING ALGORITHM

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ABSTRACT

Brain Tumor is a complex disorder attributed to irregular brain cell development. Earlier tumor identification is essential for adequate care preparation. Magnetic resonance imaging (MRI) is also recognized for being a crucial instrument for brain tumor diagnosis. MRI may be used to classify different tissues with good productivity and specificity inside the brain. In MRI, the radiation is non-ionizing, and the contrasting agents used are less toxic. Computer-Aided Diagnosis (CAD) may be almost as successful as double reading, allowing the radiologist to double vision and increase the measurement's sensitivity and precision. The proposed algorithm for brain tumor identification involves four phases: Preprocessing, Segmentation, Feature Extraction, and Classification. Noise filtering, contrast enhancement, and skull stripping are essential measures of preprocessing. Also, this paper suggests the brain MRI segmentation utilizing the K-means Cluster and Texture Pattern Matrix. Median Filter is used to eliminating noise and HF components from MRI without impacting their edges and bandwidth. Automatic Brain Tumor Prediction System Using Texture Pattern Recognition (ABPTPR) is proposed to increase brain MRI performance prediction.

Key words: MRI, Brain Tumor, Prediction, Image Matrix.

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1. INTRODUCTION

The unusual growth of cells inside or around the brain leads to a brain tumour. The WHO has grouped brain tumours into four groups. Tumors in Class II and Class I (benign), Class III and Class IV (cancerous and fatal). Earlier identification of tumours is important for effective care preparation. A brain tumour may be identified using different approaches including symptom recognition, diagnostic imaging, screening tests [1]. Proper tumour analysis can be successfully done through microscopic tissue exam (biopsy). Skull stripping is focused on connected regions and morphology. K-means clustering is applied to diagnose a brain tumor, with a segmentation method based on the texture pattern matrix (TPM). Here, the MRI segmentation algorithm's output for precision, specificity, and sensitivity is determined by manually segmented ground reality images. The research was conducted to authenticate the method's output by Area Development, Watershed, and Active Contour Model (ACM). There is also a Fuzzy C-means (FCM) algorithm introduced. The output of the segmentation algorithm is evaluated in conjunction with TPM. The parameters are used to determine segment efficiencies, such as Mean Square Error (MSE), Peak Signal Noise Ratio (PSNR), Precision, Correlation, Dice Coefficient (DC), and Jaccard Index [36].

The recovery plans in western medicine are radiation therapy, chemotherapy and surgery. Treatment effectiveness depends on tumour size, location, type and stage. Global tumour statistics show that 3.4 out of 100,000 people are affected by a malignant tumour. In [2, 3], the worldwide population affected was 2.56 per 100,000. In [30], the overall worldwide influence was 1,98 per 100,000. Around 700 people are diagnosed with a malignant tumour every day. The tumour can affect people of all ages, and certain tumours are common in children. The likelihood of forming tumours rises with age and bad lifestyle. The predicted cancer and associated disease mortality is forecast at 11.5 million in 2030 [4]. A brain tumour develops when brain cells replicate or collect more rapidly than the normal growth rate in a given brain area. The characteristics of the cells affected can shift and irregular development. Some tumours begin in a certain part of the body and spread by blood or lymph (metastasis) to other areas. The common signs of a brain tumour include pain, distorted vision, convulsions, emotional changes, fatigue, locomotive problems, and unconsciousness. Magnetic Resonance Image (MRI) is the most modern technology for the identification and diagnosis of brain tumours. MRI can distinguish between different brain tissues with good efficiency and precision. The radiation used in MRI is non-ionizing and the contrast substances used are less damaging. In MRI, radio waves, intense magnetic forces and electric field gradients are used to construct pictures of the inner bodies of the organism [31].

1.1 Brain Tumor

An abnormal brain tumor occurs causing the brain to malfunction. There are more than 100 types of brain tumours currently discovered. Normally, fresh cells are replaced as the cells in the body are old. Cells in tumours are not replaced with new cells and are abruptly multiplied. It contributes to cell aggregation and mass forming called tumour. A brain tumour is a cancer which may happen in any section of the human brain and its effects entirely depend on the location on which the brain is damaged. The reasons why most tumours are formed are not yet identified. The common risk factors Hodgson et al.[22] are ionic radiation, neurofibromatosis and vinyl chloride. Studies show that there is no connection between cell phone radiation and brain tumours [1].

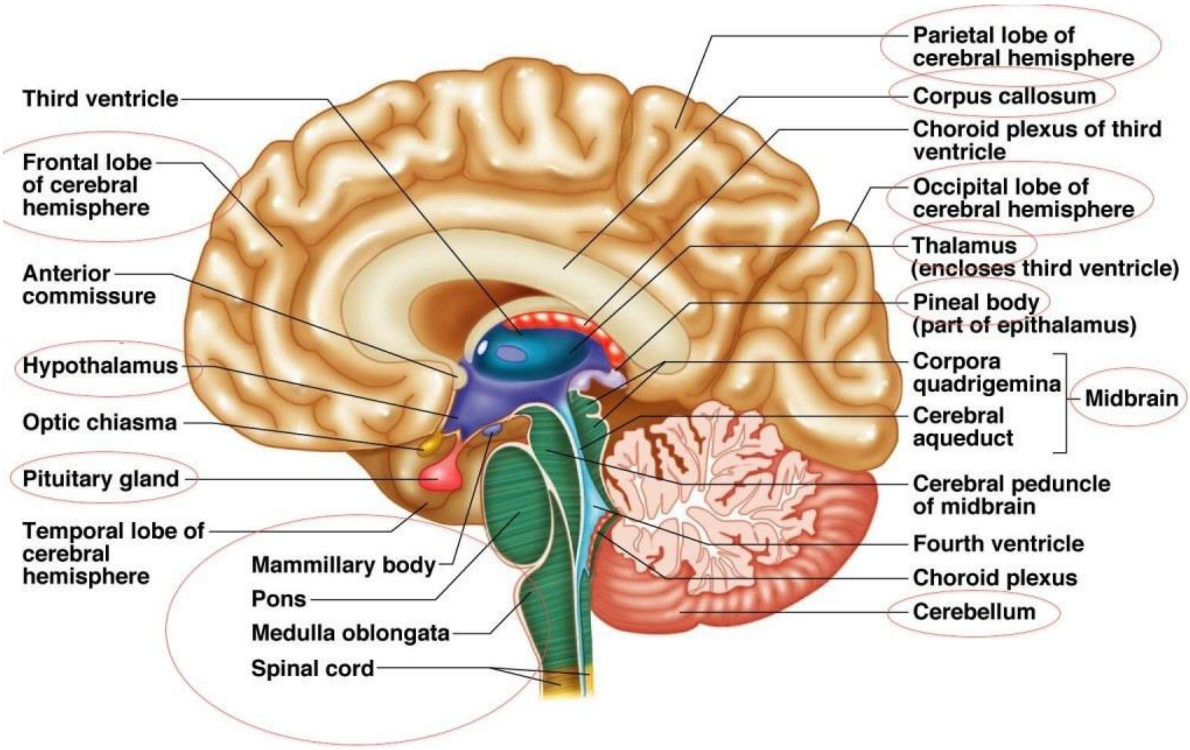

Figure 1 Anatomy of the Human Brain

However, WHO notes the mobile phone use could be carcinogenic [33]. Gliomas in continuous wireless and cellular users are at risk. When the tumour is detected in the early stages, the survival rate is high. Tumor detection is a complex process because brain tissue overlaps cover tumours

1.2 Types of Brain Tumor

There are four major brain tumour classifications. The sources, severity and cancer properties of this classification of human brain tumours are based. Primary , secondary (metastatic), malignant, and benign type of human brain tumours. Brain tumours originating in the brain are referred to as major brain tumours. It may be either malignant or benign [2013] Marella. Gliomas are typical examples of this brain tumour and come from glial cells of the brain. Brain tumours can be converted into a secondary tumour or metastatic tumour by spread from the brain to other areas of the body. Secondary (metastatic) brain tumours are detected in another region of the body as cancer cells which travel to the brain. The cancerous cells migrate by lymph or blood to enter the brain. Both metastatic tumours (cancerous) are malignant. Secondary tumours of the brain arise more frequently than core tumours. Metastatic cancer may spread through the lungs , kidneys, breasts or even skin melanomas.

Benign brain tumours are 50% of the primary brain tumours. They are slowly growing tumours with a certain form and distinct limit. Benign tumour cells look normal and don't metastasize to any other body or brain tissue. It can be life-threatening if it occurs in vital areas of the brain. Surgery is an effective way of treating a benign brain tumour. Malignant cells are brain tumours that are cancerous. They are tumours that quickly grow, invade other regions of the brain and spread to other parts of the body. A malignant tumour has a meagre survival rate. These tumours may migrate through the cerebral fluid from the brain to the spinal cord. They have no distinct borders and have an odd form.

2. RELATED WORKS

The Wiener filtering approach for noise removal in abnormal brain MRI was adopted by Paul et al. [6]. For more image processing, combinations of three separate modalities were used. In [5], Aslam et al. used a Median filter without affecting the edges for noise removal. The filter window is shifted in the picture and each pixel substitutes the median value of the next pixels. In [7] indicated homomorphic low-pass filtering to address differences in sensitivity in brain RMI. Histogram stretching has been used for intensity normalisation across all input channels. In [8], used the histogram matching method to correct the bias area. This approach aims to minimise the intervals of various patients' brain MRIs [9]. In [10] Chen et al. proposed efficient intensity standardisation in algorithms for tracking in various acquisitions and patients.

Dynamic Stochastic Resonance (DSR) was used [11] to improve the dark areas in the brain MRI. Edge sharpening was carried out using a fast and adaptive approach named Anisotropic Diffusion (AD). An accurate and computer-fast CAD method for the detection of Gliomas was introduced. The fusion scheme of the different sequence of pulses of the same modality makes the system more efficient [12, 13] for the segmentation of noisy MR images with a modified possibility FCM. Compared to the bias-corrected FCM, the misclassification error is less. The blurred membership function and standard features are weighed differently to achieve the successful segmentation of noisy pictures. Chen et al. [9] established a method for concurrently estimating the segments of the field and tissue [36].

In [14], PCA is used to segment brain MRI. He compared different changes to the PCA algorithm, and Probabilistic PCA (PPCA) is better than other methods. A mixture of PPCA and K-means offers superior precision. Yerpude et al. [36] suggested an FCM-clustered level-set segmentation algorithm and spatial restriction. The numerical complexity of FCM is solved by a level-set algorithm. In [15], adopted a probabilistic approach to improve segmentation precision. Spatial and structural problems have been solved by MRF and sparse representation. The dice coefficient obtained was greater than that of other techniques. Dong et al .developed a semi-supervised clustering method [16]. Simulated annealing is developed with the search capability. This method employed both supervised and uncontrolled clustering techniques. This method is capable of treating small clusters and finding tumour areas. Intensity characteristics for clustering were considered alone. In [17], Funmilola et al. suggested to control the noise and structural limits in Intuitionist FCM. This algorithm manages the issue of ambiguity with local knowledge. Although performance is fine, noise sensitivity is the system's flaw. In [18], developed a Genetic Algorithm (GA) based on a fluffy MRI segmentation inter cluster hostility index.

In this adaptive approach, the number of segments and the number of pixels in each segment must not be initialised. A patch-based voting system to classify brain pathologies was proposed [19]. The Region with the highest probability of tumour regions' occurrence was delineated using minimum resources and lower running time. By limiting the number of learning stages, the risk of overfitting is reduced. In [20] separated the MRI of the brain by level-set approaches. To achieve specific segments, a level range (Bayesian) has been introduced with edge detection. In this method, the volume and surface area of brain tumours were calculated accurately. In the paper [21] created a modern completely automated approach focused on texture and contour algorithms to identify and segment brain lesions. The algorithm is independent of MSI and local or global registration. Automated brain tumour detection in monospectral magnetic resonance images is a difficult task [32, 33].

A wide range of high-level features were produced to characterize the image texture. Unwanted attributes are deleted successfully. The development of a classifier signaling the presence of a tumour-affected tissue was using a machine-learning technique. An enhanced variant of the Skippy greedy snake algorithm was built to segment the tumour area with 96.8 percent precision. Hodgson et al. in [22] have developed a new approach to delineating and characterising Glioblastoma Multiforme, Metastasis, Meningioma and Brain MRI Granuloma. An integrated framework was developed to identify and extract the tumour region. Rough entropy-based thresholding was adopted in the paradigm of granular computing for the accuracy of 92 percent of the tumour area. Random Forest (RF) has been adopted, an ensemble learning method that correctly predicts the class labels of a given input by learning the training information [34].

DWT was used for the extraction of function vectors [8]. Wavelets are referred to as located base functions for extracting the coefficients used as a classifier input. The Block Processing System for effective brain RI extraction features was used [23]. The features' quality was independently evaluated, and the result was contrasted with the state of the art approaches. National statistics of tissue properties were calculated using grayscale histogram moments by Islam et al. [24]. Quantification of histogram moments helps identify shape characteristics. Multi-Scale Analysis (MSA) and Directional Spectral Distribution (DSD) were used in obtaining six-component classification function vectors [25]. These previous studies demonstrate that extraction of functions is an important phase in MRI research. The utility of derived features defines the precision of the proposed classifier. The likelihood of error will be minimised if the derived function vectors are optimum. The classifier may be efficiently applied [37].

3. METHODOLOGY: BRAIN TUMOR PREDICTION SYSTEM USING TEXTURE PATTERN RECOGNITION

In the work [26], an effective, new algorithm for the removal of the skulls using morphological operations and connected neighbours is presented. Connectivity in the neighbourhood is used to isolate brain MRI regions. The morphological operation removes unwanted regions from brain MRI in an intuitive and useful way [27]. In some scenarios in the brain region and background region the same pixel intensity appears. In this situation, it is difficult to differentiate between the true and the false context. Preprocessing improves picture efficiency by refining the picture techniques and reduces the noise. Figure 2 gives a comprehensive overview of the measures found in the suggested protocol of preprocessing [35].

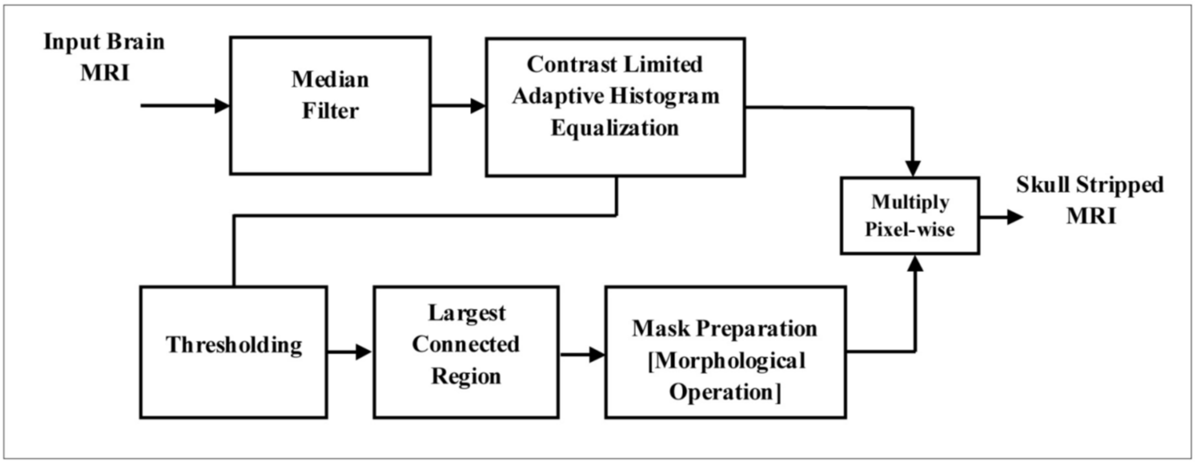

Figure 2 Block Diagram of Brain MR Image Preprocessing

Algorithm 1: Texture Pattern Recognition preprocessing

1: Get the input Brain MR image
2: 3x3 Median Filter Apply
3: Apply ABPTPR to improve MR image contrast
4: Add threshold to transform MR to binary picture
5: Find regions connected by 8-connectivity
6: Identify the area with the most linked pixels
7: Fill the largest linked Area with holes
8: Reverse the image in step 7
9: Apply dilation morphology
10: Complement the extended picture 10:
11: Multiply pixel-wise with step3 enhanced output.
12: get the brain MR picture de-noised, improved, skull extracted.

3.1 Brain MR Acquisition of Image

The technological difficulty of collecting photographs and the safety of patients renders analysis utilising true MR photographs incredibly difficult. 1.5 T scanner is used to obtain images of brain MR with consistent diagnostic quality. The ultra-light coils give a high signal-to - noise ratio and a noise reduction of 97%. It uses active and passive shimming. Thus, MR images are transferred through a Local Area Network (LAN) to the digital format computer system. In the experiment, the tranches that are suitable for the CAD system alone are used. More than 4000 photographs can be collected and transferred to digital format in less than 4 minutes. The proposed work focuses on the weighted MR images in post-contrast T1 (T1+C). Gadolinium (Gd) is injected to increase the contrast of the tumour regions. The archive consists of 48,000 pictures, 800 of which are used for training and research. The representations of the ground reality with all 800 representations are used for contrast and output measurement. It comprises of documented brain pictures and MRI of separate tumour forms (benign and malignant). The database contains pictures of 100 patients (both men and women) of the 2- to 60-year age range.

Table 1 List of Parameters used in the study

Parameters List	Parameters Description	Parameters Item
a	Echo Time	4.57 6ms
b	Image Format	DICOM
c	Inversion Time	29 ms
d	Manufacturer Name	GE Systems
e	Model Name	SignaHDxt
f	MR Acquisition Type	Cross Section 3D
g	Pixel Bandwidth	8.13 MHz
h	Series Description	T1+C
i	Slice Thickness	1.4 mm

The MR photos are moved from the database to MATLAB software throughout the image acquisition process, with an appropriate dimension and format. For neurological research, the obtained photographs should include less artefacts and disturbance. A large number of images produced during the image acquisition phase are processed. The photos are obtained in jpeg or jpg format on the CAD framework. Many photos are required to properly train and evaluate algorithms. The images are processed in 256 x 256 by design. The picture grayscale values range from 0 to 255. 0 is black, while 255 is white. A grey scale MR picture can be represented as a broad pixel intensity matrix

4. REMOVAL OF NOISE THROUGH MEDIAN FILTER PATTERN

Median Filter is a valuable method for eliminating noise and high-frequency components from MRI without disturbing their boundaries or bandwidth. The median of the adjacent pixels is determined in this approach to determine a new de-noise value for the pixel. The median is determined by sorting all pixels according to their increasing strength and choosing the centre pixel to replace the median pixel. In [28] measured median filtering as a effective method for MRI noise removal. Figure 3 demonstrates the working of a 3x3 median pattern filter.

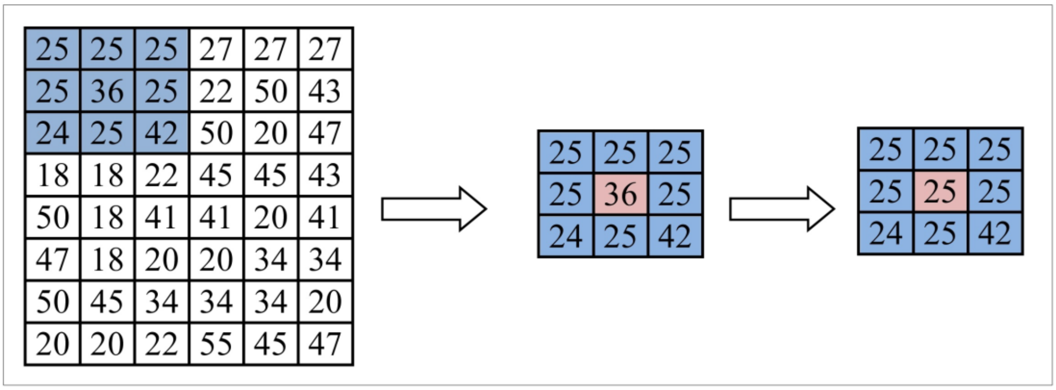

Figure 3 A 3x3 pattern median filer

Consider a 3x3 window covering 9 pixels, with 25, 25, 25, 25, 36, 25, 24, 25 and 42 intensities. Ascendingly, pixel values are rearranged including 24, 25, 25, 25, 25, 25, 25, 25, 36 and 42. The median of these pixels is 25, and the middle pixel is replaced by this pixel intensity. The philtre window is moved over the specified MR image. When the procedure is replicated, a high-resolution output is produced

5. BRAIN MR PATTERN UPGRADATION

Improvement of the picture is the method of improving the picture for a particular application. There are several algorithms and approaches used to enhance picture consistency. Relevant properties are changed to enhance the picture quality. The method will not increase or alter the quality of the picture material. It adjusts the selection of features chosen for fast identification and handling. The method of image enhancement may be carried out in the frequency or space domain. The contrast between the given MRI increases the visibility of borders and edges. The improved picture would be important for a human observer or an automated image recognition device on a machine basis. For optimizing the MR picture an advanced version of the histogram equalization procedure is used.

5.1 Equalization of histogram

A histogram can be represented as a graph which shows how much grey levels occur in a particular MR image. Some grey amounts appear more often in lower contrast videos, and some appear less or less. Equalization extends the frequency of pixel intensity levels. This method is very useful to discredit different regions of the MR picture. Histogram Equalization is a commonly used technique for improving contrast. The optimal histogram form is extracted from a Cumulative Distribution Function (CDF). The degree of amplitude is modified to prolong the peaks and to compress the troughs. Remember that the given MR picture had N pixels of L different degrees of strength. Nw is the number of intensity pixels lk. Intensity lk in the output, the likelihood density function (PDF) is Equation 1.

$$f_j(\mathrm{I_w}) = \mathrm{nw}/\mathrm{N} \tag{1}$$

The equation 2 represents the Cumulative Distribution Function (CDF) is given,

$$F_w(\mathrm{i_w}) = \sum_{l=0}^{w} f_j(\mathrm{i_l}) \tag{2}$$

The difference between the pixel intensities between adjacent artefacts in a picture can be described as contrast. The maximum intensity and minimum intensity values in an image can be regulated by a histogram. The CDF can transform the picture such that the pixel intensities are extended from maximum to minimum value by increasing the difference among intensity values throughout the histogram equalization process. This increases the contrast of the output image. The histogram of the picture before equalization is seen in Figure 4 and illustrates how new strength values are allocated during the equalization phase.

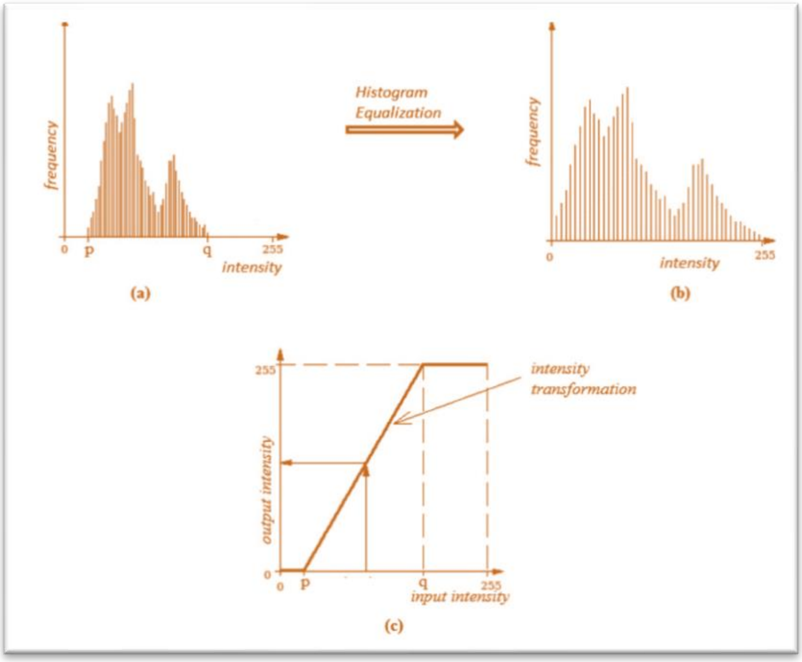

Figure 4 Equalized histogram (a) Input histogram (b) Equalized histogram(c) Graph of Intensity Transition.

5.2 Contrast Minimal Histogram Adaptive Equalization

The histogram equalisation method increases the overall picture contrast. It results in a contrast between certain regions of the picture. To prevent this issue, adaptive histogram equalisation (AHE) is used. In AHE, the histogram is divided into many histograms for a certain part of the chart. The lightest meaning is redistributed over the whole image. The local contrast is increased and the edges increased. AHE appears to intensify the noise in the frame, though. To prevent this issue, a tweaked variant of AHE called Contrast Restricted Adaptive Histogram Equalization (ABPTPR) is used. It is done by limiting the pixel intensity amplitude. ABPTPR is a tool that restricts the rise in contrast of the AHE. The pitch of the transformation function defines a given pixel's contrast amplification. The transformation function slope is exactly proportional to the CDF. The histogram is cut until computing CDF and the amplification is reduced. This lowers the CDF slopes and the transition function. Clip boundary is the value at which the histogram is cut and depends on the scale of the neighbourhood field. Regions above the clip boundary are equally redistributed to the histogram. Equation 3 provides the production amplitude of ABPTPR. Figure 5 displays the redistributed adaptive histogram.

$$Is_{out} = [Is_{\max} - Is_{\min}] \times F_w(\mathrm{Is}_{\mathrm{in}}) + \mathrm{Is}_{\min} \tag{3}$$

Where Ismax and Ismin are respectively the maximum and minimum permissible intensities. Isin is the intensity of input and Isout is the intensity of output. Fw(Ismin) is an input intensity CDF. Input intensity.

5.3 Thresholding

Thresholding is a simple method of dividing the given picture into minimum (black) and maximum (white) regions. The picture would only have two intensities (0 or 255). All other pixel intensities are transformed on a threshold value (TRS) to those two intensities. The whole pixel intensity equal to T is 255 and the remaining intensities are 0. The resultant picture is referred to as a binary image (contains just two intensities). The threshold image from the input image is determined using Equation 4.

$$F_{Trs}(\mathrm{m,n}) = \begin{cases} 1, & \text{if } \mathrm{Is_w(m,n)} \geq T \\ 0, & \text{if } \mathrm{Is_w(m,n)} < T \end{cases} \tag{4}$$

5.4 Region Selection

It is a regional mechanism in which the binary picture is separated into N-related areas. Each area is labelled with an ID number, which is called linked part marking. The resulting regions are classified as blobs. The linked pixels are neighbourhood pixels that contact the borders of a certain pixel in a 4-connectivity. The relation to the pixel of interest may be horizontal and vertical. You should bind to (x1,y) and (x, y1) a pixel at the coordinate (x , y). The linked pixels are nearby pixels that contact the edges and angles of a single pixel in 8-connectivity. The relation to the pixel of interest may be longitudinal, vertical and diagonal. You can connect a coordinate pixel (x , y) to (x1,y1). By measuring the amount of pixels used in the region, the duration of the related regions is obtained. The largest linked area is derived from these labelled regions. The neighbourhood pixel connectivity is seen in Figure-5.

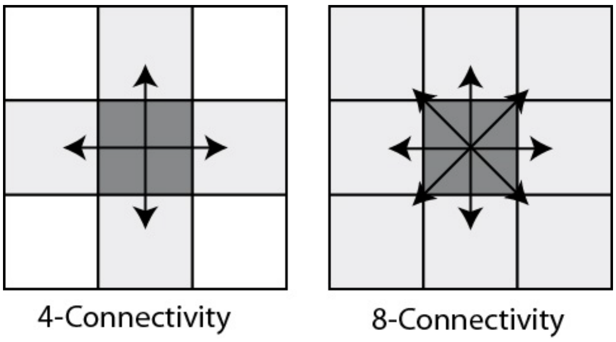

Figure 5 Pixels linked to the neighborhood utilizing 4-connectivity and 8-connectivity

It is packed with maximum density (white) pixels, considered as a boundary the largest connecting area. This gives us a white type, identical to the shape of the black brain. This photo is therefore just made up of black and white regions. The resultant picture is reversed such that the black area appears white and vice versa. In [29], dilates the picture using a 3x3 structural feature. The dilation of the image f(x , y) is done with Equation 5 using a 3x3 g(3,3) structural feature.

$$fun(x,y) \oplus gra(3,3) = \left\{ \frac{pl}{g(3,3)} \cap fun(x,y) \neq \Phi \right\} \quad (5)$$

fun (x, y)= Before dilation picture.

gra (3,3) = Dilation structural element.

g (3,3) = Structural element reflection gra(3,3).

pl= Pixel sets where gr (3,3) overlaps f (x , y).

After the dilation procedure is applied, the field in the Black Zone decreases. This picture is rotated again to obtain a white black area. Thus, the mask needed to remove the skull is acquired.

6. EXPERIMENTAL FINDINGS

For the experiment, contrast enhanced T1 weighted MR photographs of 256x256 scale collected from the database. Since all the photographs used are of the same scale, no picture resizing is required. Median Filtering is implemented for each pixel of the 3x3 window in the noise reduction step. The resulting picture is obtained. If the window size is expanded more, filtering and blurring are necessary. Figure 6 displays the picture input, the median filtered picture and its accompanying histograms.

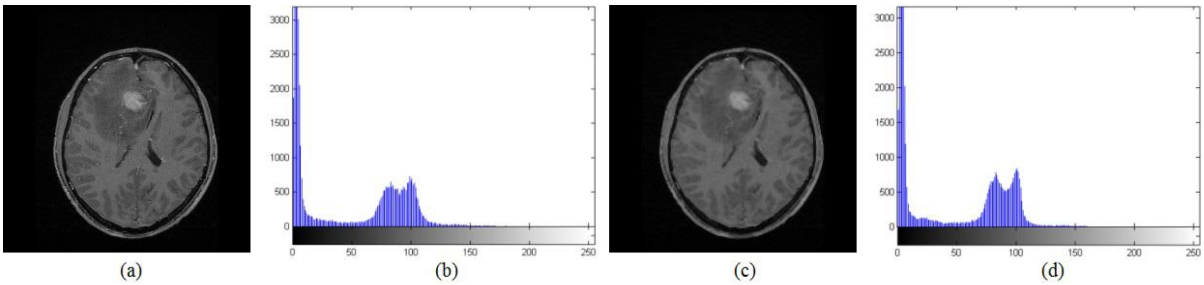

Figure 6 (a) Input Image (b) Input Image Histogram (c) 3x3 median Filteroutput(d) Filtered Picture Histogram

Median filtering takes effect until ABPTPR is implemented, ensuring that noise signals are not amplified. Figure 6 shows that the intensities are not distributed evenly over the image, which results in low contrast. ABPTPR is done via noise removed images and improved output to improve contrast. Experiments are carried out through the application of different clip limits during equalisation. The clip cap ranges from 0.00001 to 0.1 and the segmentation of each production is carried out. For clip limit 0.001 the optimum segmentation performance is obtained. The histogram is given in Figure 7 to equalise images for clip limits 0.01, 0.001, and their respective histograms. By comparing Figure 7 and Figure 6, an increase in contrast level can be understood.

When the clip cap is raised to 0.01, the contrast takes place and the picture regions can not be defined correctly. In comparison, the 0.001 clip cap offers medium enhancement, and the area is well established. So the 0.001 clip cap is well tailored to the CAD device use. The improved picture is used to create a threshold picture of binary values only. This experiment has a threshold of 40. The threshold value is obtained using the test and error method. The threshold value was optimised by two hundred photos (T=40). Linked regions are determined from the threshold picture. The linked area is given a duration for the full length. The largest connecting area is filled with holes for a white region in the dark context in the mask preparing phase. The resultant picture is reversed to achieve a black area in a white background. A structural feature of 3x3 scale extends the inverted picture to the white area. The resulting image is reversed to obtain the image of the mask. The phases in the preparation of the mask are shown in Figure 8.

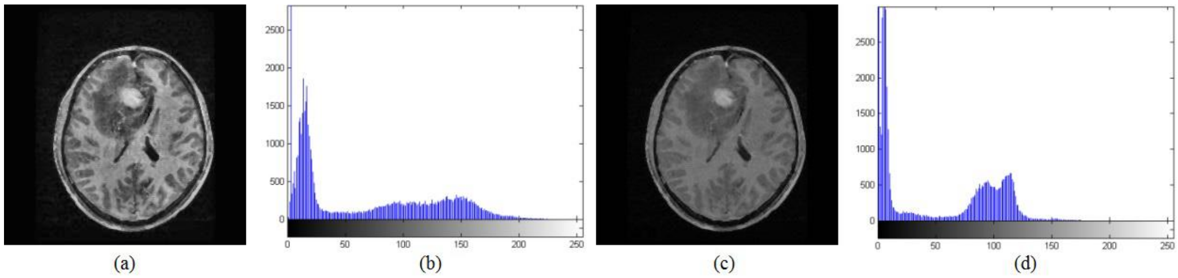

Figure 7 ABPTPR(a) Enhanced Clip Limit Image 0.01 (b) Clip Limit histogram 0.01(c) Enhanced Clip Limit Image 0.001 (d) Clip Limit histogram 0.001.

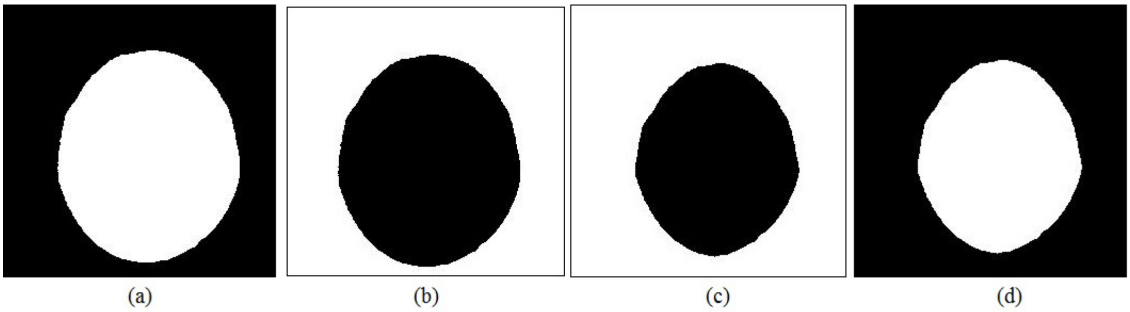

Figure 8 Mask Process Preparation (a) Largest Holes Connected Area (b)Hole Filled Complement (c) Picture dilated (d) Final Mask

Table 2 demonstrates input image contrast (Cin), processed image contrast (enhanced), contrast enhancement index (CII), PSNR, MSE, dice coefficient and jacks index for 10 sample inputs. The time of execution (TE) for each input picture is seen in the table in seconds. In this work the clip limit for improving contrast is 0.001, which avoids a problem of contrast. Contrast is enhanced at a moderate level as over-segmentation of the tumour regions can contribute to contrast. The average CII value is 1.1987. The average time for success is 0.84 seconds, suggesting lower numerical difficulty. The average accuracy of the proposed algorithm is 99,02%, compared to the manual skull stripped image. Figure 3.10 demonstrates the GUI for preprocessing algorithm outputs.

7. CONCLUSION

This paper focuses specifically on MRI preprocessing approaches such as rust reduction, enhancement of contrast and reduction of skull regions. De-noising is carried out by means of a median philtre 3x3 and is based on the ABPTPR algorithm. A new algorithm for skull stripping brain MRI is introduced, focused on mathematical anatomy and the related areas. This algorithm uses mathematical morphology's dilation property to find the regions to be removed. The precision of the algorithm for skull stripping is measured using manually stripped images. The proposed algorithm is quick because the average preprocessing time is 0.84 seconds. PSNR and MSE have an average of 76.77 dB and 0.6723. The overall Dice Similarity index value is 0.9627. The average Jaccard Index value is 0.9289. The average accuracy of the skull stripping algorithm is 99.02 percent when tested with our data set. As we focus primarily on developing a robust and accurate segmentation algorithm, we have not done any further studies on skull stretching algorithms. The precision of the proposed algorithm would improve the quality of the segmentation.

Table 2 Stripping results of Brain Skull

Image	CII	C_{in}	C_p	DICE	Jaccard	MSE	PSNR	TE
I1	1.22031	4.488435	5.21661	1.000755	0.95592	1.07205	76.692	0.884415
I2	1.24026	3.84657	4.54377	1.008105	0.96936	0.74865	78.3195	0.884835
I3	1.23333	3.527265	4.88313	1.027215	1.005375	0.3738	81.4905	0.883575
I4	1.276485	3.75396	4.56372	1.01766	0.98721	0.6069	79.2855	0.882735
I5	1.26399	3.032295	3.650325	1.018605	0.988995	0.517755	80.0205	0.883365
I6	1.274175	3.11976	3.78609	1.028265	1.00737	0.333165	82.026	0.87864
I7	1.2537	3.771285	4.523295	0.97797	0.91518	1.240575	76.02	0.891345
I8	1.23291	4.423335	5.194245	0.964845	0.892395	1.542765	75.033	0.899535
I9	1.2894	3.27852	4.026225	1.029105	1.008735	0.3003	82.488	0.89796
I10	1.30179	2.625525	3.25521	1.036665	1.023645	0.20391	84.2625	0.89586

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